

-2 / +17 lines of 13 · 1 parallel region(s) added · 4 of 7 candidate(s) kept · RESTRUCTURING (10 code line(s) added, 2 removed) · restructuring level 0

BEFORE	AFTER	file
<pre>1 #include "data.h" 2 3 real_t kernel_k23(void) 4 { 5 for (int nl = 0; nl < iterations; nl++) {</pre>	<pre>1 #include "data.h" 2 3 real_t kernel_k23(void) 4 { 5 for (int nl = 0; nl < iterations; nl++) { 6 // Capture c and d values at the start of this iteration 7 real_t c_snapshot[LEN_1D]; 8 real_t d_snapshot[LEN_1D]; 9 for (long i = 0; i < LEN_1D; i++) { 10 c_snapshot[i] = c[i]; 11 d_snapshot[i] = d[i]; 12 } 13 14 // Stage 1: Compute v (which updates w via array aliasing) 15 #pragma omp parallel for firstprivate(c_snapshot,d_snapshot) 16 for (long i = 1; i < LEN_1D; i++) { 17 v[i] = x[i] * d_snapshot[i] + c_snapshot[i]; 18 } 19 20 // Stage 2: Use w to update u 21 for (long i = 1; i < LEN_1D; i++) { 22 u[i] += w[i] * c_snapshot[i]; 23 } 24 25 dummy(a, b, c, d, e); 26 } 27 return (real_t)0; 28 }</pre>	
<pre>6 for (long i = 1; i < LEN_1D; i++) { 7 u[i] += w[i] * c[i]; 8 v[i] = x[i] * d[i] + c[i]; 9 }</pre>		
<pre>10 dummy(a, b, c, d, e); 11 } 12 return (real_t)0; 13 }</pre>		